Harika Paamu

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PROFESSIONAL SUMMARY

Data Scientist with around 4 years of experience delivering ML, NLP, and Generative AI solutions in finance, BFSI, and retail. Skilled in Python, SQL, PySpark, Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Hugging Face Transformers, LangChain, and cloud ML platforms (AWS, GCP, Azure). Experienced in building production-grade pipelines for credit risk, fraud detection, anomaly detection, and time-series forecasting on 5M+–10M+ records, with RAG architectures, vector databases, and MLOps workflows. Adept at creating interpretable models and interactive dashboards (Tableau, Power BI) to drive actionable insights, optimize operations, and reduce risk.

TECHNICAL SKILLS

Programming & Development: Python, SQL, PySpark, Pandas, NumPy, JupyterLab, FastAPI, Streamlit, H2O.ai, PyTorch, TensorFlow, ONNX , Bash/shell scripting, Java

Machine Learning & Predictive Modeling: Supervised & Unsupervised Learning, Ensemble Methods, Random Forest, Gradient Boosting (XGBoost, LightGBM, CatBoost), Time Series Forecasting, Feature Engineering, Model Evaluation & Validation, CrossValidation, Model Interpretability (SHAP, LIME)

Data Visualization & Reporting: Tableau, Power BI, Plotly, Matplotlib, Seaborn, Advanced Excel, SSRS

Artificial Intelligence & Generative AI: Large Language Models (GPT-4, LLaMA, Falcon), Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, Prompt Engineering, Embeddings & Vector Search (FAISS, Pinecone, Weaviate)

Data Engineering & MLOps: End-to-End ML Pipelines, ETL/ELT Workflow Automation, Model Deployment, CI/CD for ML, A/B Testing ,Data bricks, ADF , IBM Data Stage, Apache Kafka, API Integration

Cloud Platforms: AWS (SageMaker, EC2, S3, Lambda), GCP (Vertex AI, Google Big Query ,GCS), Azure (ADF ,Azure ML , Azure SQL) Docker, Kubernetes

Data Management: Relational Databases (PostgreSQL, MySQL), NoSQL Stores (MongoDB, DynamoDB), Data Lake & Warehouse Concepts, Snowflake, SAP HANA ,SSMS, SQL developer

PROFESSIONAL EXPERIENCE

Data Scientist, JPMorgan Chase — Dec 2024 – Present | OH, USA

- Built production-grade ML pipelines using Python, Scikit-learn, XGBoost, and NumPy for credit risk scoring and probability-ofdefault modeling across 10M+ retail banking accounts, improving risk assessment accuracy by 25% and reducing loan defaults.
- Developed NLP-driven intelligence models using BERT and spaCy to analyze customer support tickets, dispute logs, and transaction notes, cutting review and resolution time from 6 hours to under 2 hours while surfacing high-risk behaviour patterns.
- Implemented semantic search and RAG frameworks with Pinecone vector databases and transformer embeddings, indexing 500K+ historical transactions to enable sub-500ms retrieval for compliance and audit investigations, doubling analyst efficiency.
- Applied advanced machine learning techniques, including gradient boosting, ensemble models, and feature importance analysis, to detect anomalous transactions and emerging risk trends, achieving ROC-AUC of 0.91.
- Created LLM-assisted analytical workflows with LangChain and GPT architectures, enabling analysts to query structured and unstructured banking data naturally, supporting 50+ fraud and AML investigations per day.
- Established real-time model monitoring and drift detection pipelines using MLflow, FastAPI, and logging frameworks, ensuring continuous reliability of risk models and dashboards for enterprise banking operations.

Data Scientist, Hexaware Technologies — Dec 2020 – Jul 2023 | India

- Constructed large-scale predictive models using Python, Scikit-learn, XGBoost, and Pandas to analyze 5M+ financial transactions for BFSI clients, improving fraud detection precision by 28% and reducing false positives by 18%.
- Engineered time-series forecasting pipelines using Prophet and LSTM to predict customer demand and revenue trends across multiregion retail clients, reducing forecast variance by 22% and optimizing inventory planning cycles.
- Configured NLP-driven analytics solutions using Transformers and spaCy to process 1M+ customer interaction records, extracting sentiment and intent signals that improved customer retention strategies by 15%.
- Designed real-time anomaly detection frameworks using Spark, FastAPI, and AWS Lambda to monitor transaction irregularities, achieving sub-minute alert latency across distributed enterprise systems.
- Architected ensemble-based risk scoring models with LightGBM and Random Forest, integrating structured and semi-structured datasets from Snowflake and Azure Data Lake, increasing underwriting efficiency by 30%.
- Architected interactive BI dashboards using Tableau and Power BI connected to Snowflake and SQL Server, enabling CXOs to track KPIs such as churn rate, claim settlement time, and operational SLA adherence.
- Implemented end-to-end MLOps workflows using MLflow, Docker, Kubernetes, and Azure ML, automating model retraining and deployment pipelines to ensure scalability, governance, and continuous performance monitoring across cloud environments.

EDUCATION

Loyola University Chicago — Aug 2023 – May 2025 | Chicago, IL

Master of Science in Information Technology

RBVRR Women's College — Jun 2019 – May 2022 | Hyderabad, India

Bachelor of Science in Computer Science and Forensic Science

CERTIFICATIONS

Microsoft Certified: Power BI Data Analyst Associate (PL-300)
Microsoft Certified: Fabric Data Engineer Associate(DP-700)